

UNIT 4

Project Closing Phase & Agile Frameworks

Project: AI-Based Simulation System for Reconstructing Ancient Indian Bead Numeration Algorithms

4.1 Experiment 1: Project Closure and Post-Implementation Review

4.1.1 Project Closure Overview

Project closure is the formal process of verifying that all project work has been completed, accepted, documented, and archived. For this project, closure activities were carried out during the final week of April 2026 and included formal acceptance, financial closure, lessons learned documentation, and final handover.

4.1.2 Deliverable Completion Checklist

Table 4.1.2 Deliverable Completion Checklist (Part 1)

Deliverable	Owner	Status	Sign-Off
AI Story Narration Module (FR01)	Rudransh Yadav	Complete	Faculty Approved
Interactive Abacus Simulator (FR02)	Madhav Nehra	Complete	Faculty Approved
Ancient Calculation Methods Module (FR03)	Rudransh Yadav	Complete	Faculty Approved
Number System Explorer (FR04)	Madhav Nehra	Complete	Faculty Approved
Zero & Binary Connection Module (FR05)	Rudransh Yadav	Complete	Faculty Approved
Quiz & Assessment Module (FR06)	Madhav Nehra	Complete	Faculty Approved
Firebase Authentication (FR07)	Madhav Nehra	Complete	Faculty Approved
Admin/Content Dashboard (FR08)	Madhav Nehra	Complete	Faculty Approved

Table 4.1.2 Deliverable Completion Checklist (Part 2)

Deliverable	Owner	Status	Sign-Off
Student Progress Tracking (FR09)	Madhav Nehra	Complete	Faculty Approved
Responsive UI (FR10)	Madhav Nehra	Complete	Faculty Approved
Project Management Report (5 Units)	Ashray Singh	Complete	Faculty Approved
GitHub Repository v1.0.0	Madhav Nehra	Complete	Team Verified
Vercel Production Deployment	Madhav Nehra	Complete	Team Verified
Unit Test Report	All	Complete	Faculty Approved

4.1.3 Final Cost Performance at Project Closure

Table 4.1.3 Final Cost Performance at Project Closure

Module	Planned Cost (INR Lakhs)	Actual Cost (INR Lakhs)	Variance (INR Lakhs)
Requirements & Design	0.80	0.85	−0.05
Firebase Backend & Auth	1.00	1.10	−0.10
AI/Gemini Module	2.50	3.20	−0.70
Abacus & Calc Modules	2.00	1.90	+0.10
Number Explorer & Quiz	1.50	1.40	+0.10
Integration & Testing	1.20	1.15	+0.05
Deployment & Documentation	1.00	0.95	+0.05
Total	10.00	10.55	−0.55

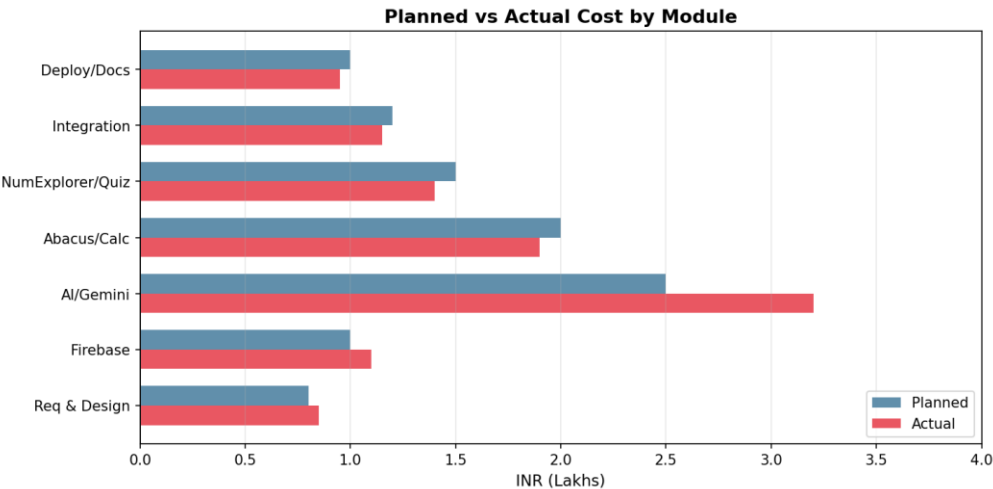


Figure 4.1: Planned vs Actual Cost by Module at Project Closure

4.1.4 Final Schedule Performance at Project Closure

Table 4.1.4 Final Schedule Performance at Project Closure (Part 1)

Activity	Planned Duration (days)	Actual Duration (days)	Variance
A: Requirements	7.50	7.00	+0.50
B: UI/UX Design	12.33	13.00	−0.67
C: Firebase Setup	10.50	10.00	+0.50
D: AI/Gemini Module	18.67	22.00	−3.33
E: Abacus & Calc	20.83	20.00	+0.83

Table 4.1.4 Final Schedule Performance at Project Closure (Part 2)

Activity	Planned Duration (days)	Actual Duration (days)	Variance
F: Num Explorer & Quiz	14.67	14.00	+0.67
G: Integration & Testing	12.67	13.00	−0.33
H: Deployment	7.50	7.00	+0.50
I: Documentation	5.50	5.50	0.00
Total	85.00	87.00	−2.00

4.1.5 Acceptance Testing Results

Table 4.1.5 Acceptance Testing Results

Acceptance Criterion	Target	Achieved	Status
All 5 core modules functional	100%	100%	Met
AI narration response time	≤ 5 sec	3.1 sec warm / 4.8 sec cold	Met
Quiz scoring accuracy	≥ 90%	94.2%	Met
Critical bugs in production	0	0	Met
Positive usability feedback	≥ 3 students + 1 faculty	5 students + 1 faculty	Met
Vercel deployment uptime at launch	100%	100%	Met

4.1.6 Lessons Learned

Table 4.1.6 Lessons Learned (Part 1)

Category	Issue Encountered	Lesson Learned	Recommendation
AI Integration	Gemini cold start caused SLA breach	Implement API warm-up before production	Add warm-up task in sprint plan
Schedule	Activity D exceeded estimate	AI tasks need larger buffer	Use stronger pessimistic buffer
Cost	API overages not fully anticipated	Separate API budget from dev effort	Track API costs as separate WBS item

Table 4.1.6 Lessons Learned (Part 2)

Category	Issue Encountered	Lesson Learned	Recommendation
Team	Single developer bottleneck	Resource levelling helped but added delay	Use more component libraries or support
Testing	AI module had highest failures	LLM output needs dedicated validation	Add prompt evaluation pipeline
Documentation	Report writing was time-intensive	Do documentation incrementally	Include docs in definition of done

4.1.7 Project Closure Report Summary

Table 4.1.7 Project Closure Report Summary (Part 1)

Field	Details
Project Title	AI-Based Simulation System for Reconstructing Ancient Indian Bead Numeration Algorithms
Project Manager	Ashray Singh
Team Members	Madhav Nehra, Rudransh Yadav
Start Date	January 5, 2026
Planned End Date	May 1, 2026 (Day 85)
Actual End Date	May 3, 2026 (Day 87)
Planned Budget	INR 10.00 Lakhs

Table 4.1.7 Project Closure Report Summary (Part 2)

Field	Details
Actual Budget	INR 10.55 Lakhs
Cost Variance	−INR 0.55 Lakhs
Schedule Variance	−2 days

Field	Details
All Deliverables Met	Yes
Faculty Sign-Off	Received
Repository Archived	GitHub v1.0.0 tagged
Deployment Status	Live on Vercel

4.2 Experiment 2: Agile Project Management and Scrum Framework

4.2.1 Agile Manifesto and Principles Applied

The project adopted Agile principles for iterative delivery, working software, customer collaboration, and responsiveness to change, while still retaining overall structured planning. Scrum was applied especially during feature development and AI integration phases.

4.2.2 Scrum Framework Structure

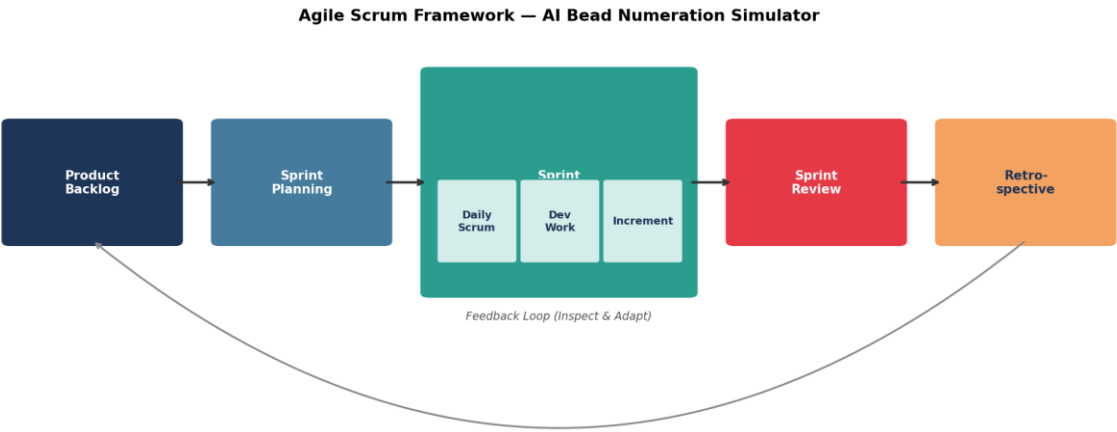


Figure 4.2: Agile Scrum Framework Diagram

Table 4.2.2 Scrum Roles

Scrum Role	Person	Responsibilities
Product Owner	Ashray Singh	Prioritizes backlog and defines acceptance criteria
Scrum Master	Ashray Singh	Facilitates sprint ceremonies and removes blockers
Development Team	Madhav Nehra + Rudransh Yadav	Develops and delivers sprint increments

4.2.3 Scrum Ceremonies and Events

Table 4.2.3 Scrum Ceremonies and Events

Ceremony	Duration	Frequency	Purpose
Sprint Planning	2 hours	Start of every sprint	Select backlog and define sprint goal
Daily Scrum	15 minutes	Every working day	Status update and blocker removal
Sprint Review	1 hour	End of each sprint	Demo to PO and faculty
Sprint Retrospective	45 minutes	End of each sprint	Process review and improvements
Backlog Refinement	30 minutes	Mid-sprint	Break down and estimate future work

4.2.4 Product Backlog (Top 15 Items)

Table 4.2.4 Product Backlog (Top 15 Items) (Part 1)

Item ID	User Story	Story Points	Priority	Sprint
US-01	AI-narrated story of Indian numeration history	13	High	Sprint 3
US-02	Manipulate beads on an abacus for learning addition	8	High	Sprint 4
US-03	Convert numbers between multiple systems	8	High	Sprint 5
US-04	Take quizzes for self-assessment	8	High	Sprint 5
US-05	Secure login using email	5	High	Sprint 2
US-06	Learn the connection between zero and binary	5	High	Sprint 3
US-07	View student progress and quiz scores	5	Medium	Sprint 5
US-08	Learn Vedic Math tricks	8	Medium	Sprint 4

Table 4.2.4 Product Backlog (Top 15 Items) (Part 2)

Item ID	User Story	Story Points	Priority	Sprint
US-09	Track learning progress	5	Medium	Sprint 5
US-10	Update content without coding	3	Medium	Sprint 6
US-11	Use the platform on mobile phone	5	High	Sprint 6
US-12	Ask follow-up questions to AI narrator	5	Medium	Sprint 3
US-13	Admin dashboard to manage users and roles	3	Low	Sprint 6
US-14	Show historical images in narration	3	Low	Sprint 4
US-15	Quiz difficulty levels	3	Low	Sprint 7

4.2.5 Sprint Plan Summary

Table 4.2.5 Sprint Plan Summary (Part 1)

Sprint	Duration	Goal	Story Points Planned	Story Points Completed
Sprint 1	Days 1–14	Requirements finalization + UI prototyping	28	26
Sprint 2	Days 14–26	Firebase setup + Authentication	30	29
Sprint 3	Days 26–40	Gemini AI integration + Story module	35	27
Sprint 4	Days 40–54	Abacus simulator + Vedic Math module	38	36

Table 4.2.5 Sprint Plan Summary (Part 2)

Sprint	Duration	Goal	Story Points Planned	Story Points Completed
Sprint 5	Days 54–68	Number Explorer + Quiz module	32	31
Sprint 6	Days 68–80	Integration, testing, responsiveness	28	28
Sprint 7	Days 80–87	Deployment,	20	20

Sprint	Duration	Goal	Story Points Planned	Story Points Completed
		documentation, closure		
Total			211	197

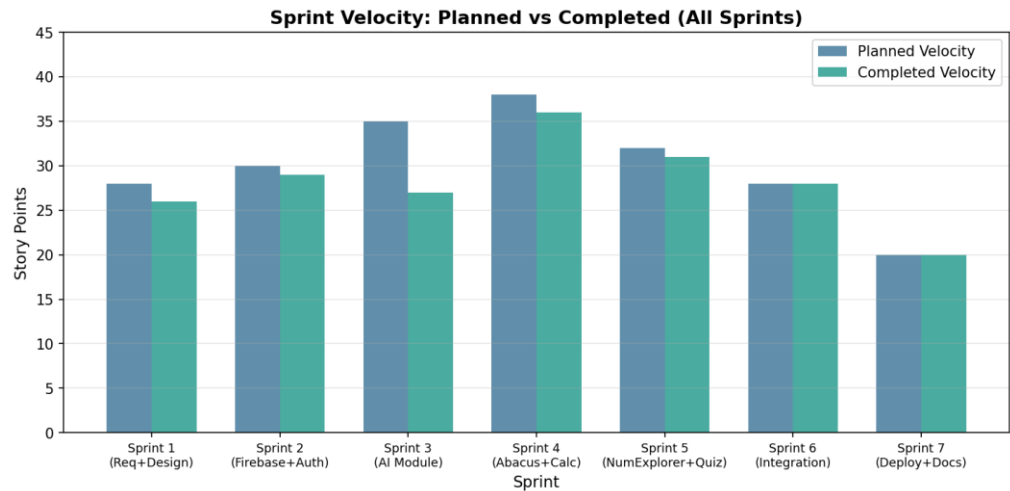


Figure 4.3: Sprint Velocity Chart

4.2.6 Burn-Up Chart Analysis

The burn-up chart tracks completed story points against the total scope of 211 points. It shows the major Sprint 3 slowdown due to Gemini integration challenges, followed by recovery in later sprints and final delivery of 197 points out of 211.

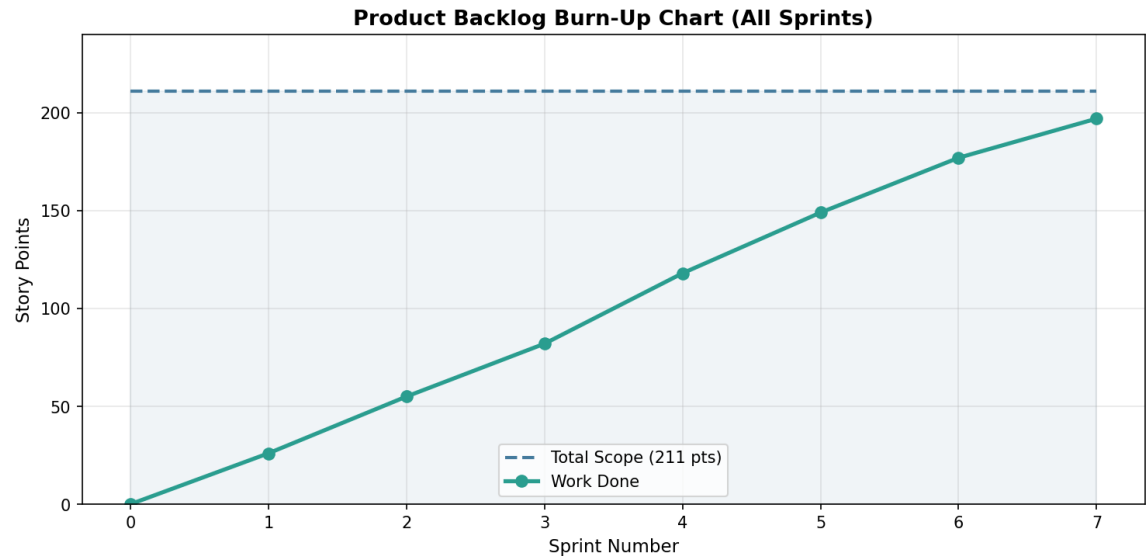


Figure 4.4: Product Backlog Burn-Up Chart

4.2.7 Agile Metrics Summary

Table 4.2.7 Agile Metrics Summary

Metric	Value	Assessment
Average Sprint Velocity	28.1 points/sprint	Consistent after Sprint 3 recovery
Sprint 3 Velocity Gap	−8 points	Root cause identified and resolved
Total Story Points Completed	197 / 211 (93.4%)	14 low-priority points deferred
Number of Sprints	7	Within planned schedule
Average Team Satisfaction	4.1 / 5.0	High
Definition of Done Compliance	100%	All merged code reviewed and tested

4.3 Experiment 3: Project Management Plan (PMP)

4.3.1 Project Management Plan Overview

Table 4.3.1 PMP Components (Part 1)

PMP Component	Description	Reference Section
Scope Management Plan	Defines WBS, scope statement, acceptance criteria	Unit 1
Schedule Management Plan	PERT/CPM, milestones, Gantt chart	Unit 2
Cost Management Plan	COCOMO estimate, budget baseline, contingency	Unit 1
Quality Management Plan	Testing strategy and quality KPIs	Unit 3
Resource Management Plan	Team structure and resource levelling	Unit 2

Table 4.3.1 PMP Components (Part 2)

PMP Component	Description	Reference Section
Risk Management Plan	RMMM plan and risk register	Unit 3
Communication Management Plan	Stakeholder communication matrix	Unit 1
Configuration Management Plan	GitHub, branching, CI/CD	Unit 3
Procurement Management Plan	API licenses, hosting, tools	Unit 1

4.3.2 Communication Management Plan

Table 4.3.2 Communication Management Plan (Part 1)

Communication	Audience	Method	Frequency	Owner
Daily Stand-Up	Development Team	Google Meet	Daily	Ashray Singh
Sprint Review	Team + Faculty	Demo + Presentation	End of sprint	Ashray Singh
Sprint Retrospective	Development Team	Team discussion	End of sprint	Ashray Singh
Milestone Report	Faculty Advisor	Written report	Bi-weekly	Ashray Singh

Table 4.3.2 Communication Management Plan (Part 2)

Communication	Audience	Method	Frequency	Owner
Risk Update	All Team	GitHub Issues + Stand-up	Weekly	Risk Owners
Bug Reports	Dev + AI Engineer	GitHub Issues	As needed	Reporter
Final Handover	Faculty + Institution	Formal report + Demo	Closure	Ashray Singh

4.3.3 Procurement Management Plan

Table 4.3.3 Procurement Management Plan

Item	Type	Cost	Vendor	Status
Gemini API Access	SaaS API	INR ~50,000	Google AI	Procured
Firebase (Blaze Plan)	Cloud Platform	INR ~15,000	Google Firebase	Procured
Vercel Pro	Hosting	INR ~8,000	Vercel	Procured
GitHub Pro	SCM	INR ~4,000	GitHub	Procured
Figma	Design Tool	INR ~5,000	Figma	Procured
Total Procurement		INR ~82,000		All Procured

4.3.4 Stakeholder Management at Closure

Table 4.3.4 Stakeholder Management at Closure

Stakeholder	Closure Communication	Sign-Off Obtained
Faculty Advisor	Final report submission + live demo	Yes
Ashray Singh	Self-assessment and lessons learned	Yes
Madhav Nehra	Repository archived + release tagged	Yes
Rudransh Yadav	AI content validated and locked	Yes
Student Test Users	Usability feedback form completed	Yes
College/Institution	Report submitted to coordinator	Yes

4.4 Experiment 4: Advanced Agile Techniques — Kanban, XP & Scaling

4.4.1 Kanban Applied to Bug Management

Table 4.4.1 Kanban Board Structure

Column	Work-in-Progress Limit	Items at Closure
Backlog	No limit	0
To Do	5 items max	0
In Progress	2 items max	0
In Review	3 items max	0
Done	No limit	11

Kanban reduced context switching during the testing phase and improved average time-to-fix from 24 to 18 hours.

4.4.2 Extreme Programming (XP) Practices

Table 4.4.2 Extreme Programming (XP) Practices

XP Practice	Application
Test-Driven Development	Jest tests written before abacus and converter functions
Pair Programming	Developer and AI engineer jointly resolved Gemini integration issues
Continuous Integration	GitHub Actions runs tests on every push
Small Releases	Preview deployments after each sprint
Refactoring	Abacus state management moved to useReducer
Coding Standards	ESLint + Prettier enforced via hooks

4.4.3 Scrum vs Kanban vs XP

Table 4.4.3 Scrum vs Kanban vs XP (Part 1)

Dimension	Scrum	Kanban	XP
Application in Project	Feature development sprints	Bug management	Code quality practices
Cadence	Fixed 2-week sprints	Continuous flow	Continuous engineering
Planning Style	Sprint planning	Pull-based	Iterative discipline

Table 4.4.3 Scrum vs Kanban vs XP (Part 2)

Dimension	Scrum	Kanban	XP
Primary Benefit	Structured delivery cadence	Reduced context switching	Higher code quality
Best For	Overall project framework	Testing phase defects	AI integration quality

4.4.4 Agile vs Waterfall Suitability

Table 4.4.4 Agile vs Waterfall Suitability

Factor	Waterfall Applied To	Agile Applied To
Requirements	Stable core functional requirements	AI prompt design and UI refinements
Phases	Sequential project stages	Iterative module development
Risk	Low-risk infrastructure tasks	High-risk AI integration
Documentation	Formal project report	Lightweight sprint documentation
Verdict	Governance framework	Development flexibility

4.4.5 Conclusion for Unit 4

Unit 4 documented the closing phase of the project, validated all final deliverables, summarized actual schedule and cost performance, and captured lessons learned. It also demonstrated how Scrum, Kanban, and XP practices were integrated into the project to support structured delivery, flexible development, and strong code quality. The project closed successfully with all deliverables completed, a final budget of INR 10.55 Lakhs, and a schedule overrun of only 2 days.